What Do Document Forgery Localizers Learn from Box Masks? Ground-Truth Geometry, a Hidden Marker Shortcut, and Box Detection on AI-Inpainted Financial Documents

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***Abstract*—Benchmarks for AI-inpainted document forgery, such as AIForge-Doc, annotate each edit with the bounding box of the rewritten field. We ask what such labels teach a localizer. Using the authentic source images released with the training forgeries, we derive pixel-change and glyph masks and train the same network on box, pixel-change and glyph masks, changing nothing else. Under its own box metric the box-trained U-Net scores 0.92 pixel AUPRC; against the pixels that actually changed the same predictions score 0.37, and each ground truth ranks its own model first. A second decoder (DeepLabV3+) reproduces the effect. We find that a 1–3 px residue of the green edit marker drawn by the generation pipeline survives in over 80 % of Gemini forgeries: removing it lowers the box model's per-image AP from 0.99 to 0.77, and pasting it alone onto an authentic image is enough to light up the field. Retraining on marker-repaired images (one seed) removes this dependence while keeping most of the box score. A tiled YOLO11 detector trained on the same boxes is the best model under the box metric (0.97 test AUPRC), depends less on the marker and transfers better to Ideogram edits (0.87 vs. 0.63 AP), yet reaches only 0.37 validation AUPRC under the pixel-change truth. Box scores on this benchmark therefore measure field detection, not pixel-level tamper localization; we recommend reporting both, auditing generation residue, and releasing derived masks and probes, which we provide.**

***Index Terms*—document forgery, image forensics, tamper localization, ground truth, shortcut learning, AI inpainting, object detection**

I. Introduction

Image-editing services built on multimodal generators can now rewrite a single amount on a receipt or a field on a form without a visible seam. Detectors trained on classical splicing or on synthetic document tampering transfer poorly to this setting: on AIForge-Doc v1 [1], the generic localizer TruFor [2] reaches a pixel IoU of 0.36 zero-shot and document-specific models fall further, and on its successor humans perform at chance [3]. The obvious response is to train localizers on these benchmarks. We ask a prior question: *what does the ground truth of such a benchmark actually teach a localizer, and what do its scores measure?*

AIForge-Doc annotates every forgery with the bounding box of the edited field. Our audit of the released files shows that every mask is a rectangle, that only a median 49 % of the box pixels differ from the paired authentic image, and that one third of the forgeries also change pixels outside the box. A segmentation network trained on such masks is rewarded for a rectangle around any detectable cue, which is the setting in which shortcut learning thrives [4]. Annotation geometry has been questioned for natural-image edits [5], but not for AI-inpainted documents and not with controlled training. AIForge-Doc releases the authentic source of every training forgery, which lets us derive a pixel-change ground truth and change *only* the training mask.

Our contributions are:

1) **A dual-ground-truth protocol.** The same localizer trained on box, pixel-change and glyph masks (three seeds each) is scored under all three; the box-trained model's 0.92 validation AUPRC drops to 0.37 under the pixel-change truth and each ground truth ranks its own model first (Sec. V-A). The effect replicates with a second decoder (DeepLabV3+).
2) **A hidden marker shortcut, removed and re-measured.** The generation pipeline leaves a 1–3 px green line of the drawn edit marker in over 80 % of Gemini forgeries. Pasting only this line onto an authentic image makes the box model "find" the field; we retrain every condition on marker-repaired images (one seed) and report how much of the benchmark score survives (Sec. V-C).
3) **A region-appearance decomposition** that separates the content change from the generator's re-rendering of the field (Sec. V-D).
4) **Box detection versus segmentation.** A tiled YOLO11 detector trained on the same boxes, scored with the same code under both ground truths and with box AP, beats every segmenter under the box metric (test AUPRC

0.969), relies less on the marker and transfers better across generators, yet scores like the box U-Net under the pixel-change truth (Sec. V-G).

5) **Negative results** on the paste-back seam, a global re-encoding trace, held-out-generator transfer with pixel supervision (three seeds), longer training and test-time enhancement.

We release the audit, derived masks, controls, probes and evaluation code; the data themselves are not redistributed.

II. Related Work

A. Document and text tampering localization

Most document forensics work treats localization as dense prediction. DocTamper [6] paired a large synthetic corpus with a network that reads JPEG block-DCT statistics, and later localizers refine how frequency and RGB cues are fused or made robust to recompression [7]–[14]. A parallel line, originating in scene text, scores whole text instances instead of pixels: Tampered-IC13 adapts scene-text detectors so that each detected instance is also judged genuine or tampered [15], [16]. Text-centric benchmarks for receipts and documents, including T-SROIE, the ICDAR 2023 competition, RTM and a recent human-crafted real-world set, broaden the data, with labels that range from instance boxes to pixel masks [17]–[20]. Newer systems couple localization with explanation through multimodal language models or generative restoration [21]–[25]. For API-inpainted financial documents, AIForge-Doc v1 and v2 [1], [3] and related studies [26]–[30] report detectors in a zero-shot setting only. Two non-archival projects train on these data: one reports a process-twin control on v2 [31] and one a tile-level page classifier [32]; neither studies pixel-level ground truth.

B. Ground-truth semantics and dataset shortcuts in forensics

A network optimizes whatever signal its labels reward, which need not be the intended evidence [4]. In forensics this has been shown for JPEG grid alignment [8], [10], for trivial artefacts of benchmark construction [33], [34], and for frequency bias [35]. Inpainting studies report that whole-image re-decoding defeats splicing localizers while handing image-level detectors a global cue unrelated to the edit [36], [37], motivating generator-fingerprint localizers [38]; they also find that fine-tuned localizers acquire an object prior once masks are semantic [39], and that small edits remain the hardest case [40], [41]. Closest to our question, Shang et al. [5] argue that object-shaped masks label many unchanged pixels and miss real edits, so mask-only scores can rank detectors wrongly; on natural images, audits likewise question what pixel agreement measures [42] and separate the shifts behind poor generalization [43]. In segmentation, box labels are a well-known form of weak supervision that must be refined before they approximate object extent [44], [45], and metric choice should follow the structure of the target [46]. We examine the same issue for box-annotated document forgeries, where the box is also the paste-back region.

C. Detection versus segmentation for small regions

Manipulation has been localized with region-proposal detectors since RGB-N [47], and single-stage detectors [48], now available in maintained form as YOLO11 [49], predict boxes directly and are scored with COCO-style average precision [50]. When the forensic label is a field box, a detector matches the annotation geometry by construction, whereas encoder–decoder segmenters such as U-Net and DeepLabV3+ [51], [52] must spread a box label over pixels that may be unchanged. Edited fields in receipts are also small relative to the page; small objects are a recognized weakness of detectors, especially after downsampling [53], and slicing the image into overlapping tiles at native resolution is a standard remedy [54]. Multi-task models that predict boxes and masks jointly [55], and recent forgery models that couple image-level detection with pixel localization [56], suggest that box and pixel supervision can be combined. To our knowledge, no prior study compares a box detector and a segmenter on the same AI-inpainted document benchmark while varying the semantics of the ground truth used for training and scoring.

III. Dataset and Audit

AIForge-Doc v1 [1] contains 4,061 forged PNG images (3,249 train / 812 test) derived from CORD, WildReceipt, SROIE and XFUND; receipts form 89.7 % and forms 10.3 %. Each image has one edited field, produced by Gemini 2.5 Flash Image (89.6 %) or Ideogram v2 Edit (10.4 %), and a binary mask. We recomputed all statistics from the files.

Mask semantics. All masks are single axis-aligned rectangles. With $D = \max_c |I^f_c - I^a_c|$ the per-pixel difference between a forgery I^f and its authentic source I^a (3,247 of 3,249 training pairs verified by perceptual hash and pixel difference), the median fraction of box pixels with $D > 8$ is 0.49 (IQR 0.37–0.62), and 33.5 % of pairs also change pixels outside the box (Gemini 35.9 %, Ideogram 11.9 %), typically by re-rendering the whole text line. Half of the Gemini outputs carry a global $\pm$1–3 re-encoding difference that disappears at $\varepsilon \geq 4$.

Scale. The median edited area is 0.98 % of the image (50.7 % of images below 1 %); the median box height is 33 px while images span 0.03–35 MP, so all models run at native resolution with tiling.

The marker residue. For Gemini, the dataset paper draws a 3-px bright-green rectangle on the context crop to mark the edit region [1] (Sec. 3.4 there). A saturated-green rule ($G - \max(R, B) > 50$, $G > 110$) restricted to a 6-px ring around the box and absent from the authentic image finds residue of this line ($\geq$20 px) in 83.5 % of the 2,466 Gemini and 1.4 % of the 294 Ideogram forgeries of the training split (no false alarm on 600 authentic images). Test forgeries mostly lack a paired authentic image, so for subset analyses only we label a test image as *with residue* if any saturated-green pixel lies in the 6-px box ring (86 % of Gemini, 2 % of Ideogram forgeries over the whole dataset); model inputs are never altered by this label. Marker presence and generator are therefore strongly

confounded (Cramér's $V = 0.61$), more than generator and source ($V = 0.43$; WildReceipt has no Ideogram edit).

Splits and negatives. The official test split is untouched. Training images are split by source-document groups into 2,762 train and 487 validation images with no group overlap; tool, source, area and marker distributions do not differ across the three splits (χ^2 $p \geq 0.30$). Of the test forgeries 8.9 % share a source document with training; we report the 740 leakage-free images separately. The released test "authentic" folder is not the mirror of the test forgeries (86 matches), so image-level detection is evaluated on validation pairs only. Fourteen degenerate training pairs (forged $\approx$ authentic, or all changes outside the box) are excluded from training (2,748 images remain). We redistribute only identifiers and derived masks (CC BY-NC-SA 4.0).

IV. Method

A. Derived ground truths

From every verified pair we derive, besides the official BOX mask M^b: PIX, the pixel-change mask $\{D > 8\}$ cleaned by a 3×3 opening, removal of components below 8 px and a 3×3 closing; and GLYPH, Sauvola text pixels [57] of $I^f \cup I^a$ intersected with a 7×7 dilation of $\{D > 8\}$, with polarity inverted on dark backgrounds. Median IoU with the box is 0.35 (PIX) and 0.27 (GLYPH). Because the marker line differs from the authentic image, it is part of PIX (median 10 % of its positives in affected pairs); the variants PIX_{nm} and GLYPH_{nm} remove the dilated residue pixels.

B. Controls and probes

On the validation pairs we build: *benign paste-back* (box region of the authentic image up/down-sampled $2\times$ and pasted back, i.e. compositing without content change) and *full-image resampling* [36]. Input-level probes (Fig. 1) are: *forged-repaired* (I^f with residue pixels copied from I^a); *authentic+marker* (I^a with only the residue pixels from I^f); *render-only* (I^f with the dilated PIX region and the residue reverted to I^a, so no character changes but the generator's sub-threshold re-rendering of the field remains); *content-only* (I^a with only the dilated PIX_{nm} region taken from I^f, so characters change but the surrounding re-rendering and the marker are absent); a boundary/placebo/interior blur ablation of the box; and test-time degradations (JPEG Q75, $0.75\times$ rescale, Gaussian σ=1, CLAHE, grayscale).

C. Models

Segmenters. A U-Net [51] with an ImageNet-initialised ResNet-18 encoder [58] (14.3 M parameters) is trained on 512×512 native-resolution crops (70 % centred on training-mask positives with jitter, 30 % random), BCE+Dice loss [59], AdamW [60] (10^{-4}, cosine), batch 16, bf16, 10 epochs, model selection by validation AUPRC under the *training* mask. Inference is full-image sliding window (512, stride 448); the 44 images whose longer side exceeds 4096 px are downscaled to 4096 for inference and the map is upsampled back. The training mask is the only factor varied (BOX, PIX, GLYPH; seeds 42–44). One-factor extensions, seed 42: (i) DeepLabV3+ [52] with the same encoder; (ii) training on marker-repaired images with BOX, PIX_{nm}, GLYPH_{nm}; (iii) a joint model with a second output channel supervised by BOX (loss $\mathcal{L}_{\text{pix}} + 0.5\,\mathcal{L}_{\text{box}}$, evaluation on the pixel channel); (iv) 20 epochs; (v) Gemini-only training for the held-out-generator test (seeds 42–44).

Box detector. YOLO11 [49] (single class) is trained on the same BOX boxes. Because the median field is 33 px high, images are cut at native resolution into 1024×1024 tiles (stride 768; every tile holding $\geq$ 50 % of the box is labelled with the clipped box, and one other random tile per image is kept with an empty label, which can contain a sliver of the field) and inference is tiled with cross-tile NMS, following slicing-aided detection [54]. Augmentation is conservative (no mosaic or scale jitter, which would add resampling traces); AdamW 10^{-3}, batch 8, 30 epochs, patience 10, checkpoint selection by Ultralytics' default fitness ($0.1\,\text{mAP@0.5} + 0.9\,\text{mAP@[.5:.95]}$) on validation tiles. Boxes are taken from the released metadata, which agrees with the mask extent except for 12 images (0.3 %). One-factor ablations (seed 42): whole image resized to 1280; default Ultralytics augmentation; one extra empty authentic tile per image; model size n/s/m; Gemini-only training. For pixel metrics, each box paints its confidence into a map (maximum over overlapping boxes), so that detector and segmenters pass through *identical* evaluation code.

D. Evaluation and statistics

Every model is scored against BOX, PIX and GLYPH on the 487 validation images and against BOX on the 812 test images (PIX where a test pair exists, n=89). Metrics are pooled pixel AUPRC (exact, from 256-bin histograms over all pixels), per-image AP, F1/IoU at a threshold τ^* calibrated per model and ground truth on validation, and, for all models, box-level AP@0.5 and AP@[.5:.95] (COCO-style [50]; segmenter boxes from connected components at τ^*, scored by their maximum probability). Image-level discrimination compares forged with authentic validation images using the 99th-percentile score (benign controls) or the maximum score (input probes, Tables II and III). Uncertainty is a 1,000-sample bootstrap over source-document groups [61]; contrasts within a hypothesis family are Holm-corrected [62]. Four hypotheses and their falsification criteria were fixed before the full-data runs: H1, the training ground truth changes scores and model ranking; H2, the box model relies on the box boundary or rectangle shape; H3, pixel or glyph supervision transfers better to an unseen generator and degradations; H4, forgeries carry a global trace outside the box; the marker, render/content, YOLO and joint-loss analyses were added afterwards and are reported as exploratory.

V. Results

A. H1: the ground truth decides the ranking

Table I scores every model under every ground truth. With the U-Net, the BOX-trained model reaches a pooled AUPRC

TABLE I

POOLED PIXEL AUPRC ON VALIDATION (487 IMAGES) FOR EACH TRAINING MASK (ROWS) SCORED AGAINST EACH GROUND TRUTH (COLUMNS); MEAN±SD OVER THREE SEEDS WHERE AVAILABLE. BOLD: BEST PER COLUMN WITHIN AN ARCHITECTURE.

Model	Train	BOX	PIX	GLYPH
U-Net (3 seeds)	BOX	**0.923±.015**	0.374±.008	0.257±.007
	PIX	0.579±.014	**0.610±.012**	0.511±.010
	GLYPH	0.351±.015	0.379±.012	**0.676±.019**
DeepLabV3+	BOX	**0.939**	0.382	0.256
	PIX	0.537	**0.583**	0.432
	GLYPH	0.345	0.401	**0.549**
YOLO11s (3 seeds)	BOX	**0.965±.003**	0.368±.015	0.221±.003
U-Net, marker-repaired	BOX	0.916	0.389	0.266
	PIX_{nm}	0.513	0.544	0.455
	GLYPH_{nm}	0.348	0.371	0.661
U-Net + box aux.	PIX	0.647	0.622	0.518
	GLYPH	0.370	0.391	0.636
U-Net, 20 epochs	BOX	0.926	0.373	0.259
	PIX	0.568	0.594	0.488
	GLYPH	0.398	0.390	0.645

of 0.923 under BOX but 0.374 under PIX and 0.257 under GLYPH; on the official test set the same predictions give 0.950 under BOX and 0.427 under PIX (n=89 pairs). Each ground truth ranks the model trained on it first, and the BOX ranking is the reverse of the GLYPH ranking. DeepLabV3+ reproduces the pattern with nearly identical numbers (0.939 $\rightarrow$ 0.382), so the effect is a property of the supervision, not of the U-Net decoder. Under its own metric the BOX model is the best localizer in the study; under a pixel-change metric it is worse than a model that never saw a box. Most of the 0.92 is therefore earned by labelling unchanged pixels inside the rectangle.

B. Shortcut and trace controls (H2, H4)

The BOX model predicts rectangles (median rectangularity 0.87 vs. 0.32 for PIX and 0.05 for GLYPH) and is indifferent to blurring the box interior (relative AUPRC change $+1.2\,\%$) while a 5-px blur of the box rim costs $-2.0\,\%$ (placebo ring 0.0 %; three seeds): it uses the field's extent more than its content, but the boundary seam is only a small cue (H2 partly supported). A ResNet-18 trained on 224-px crops that never touch the box separates forged from paired authentic images at chance (AUROC 0.516; 0.980 for box-centred crops through the same pipeline), so no global re-encoding trace is exploitable (H4 rejected). Benign paste-back and full-image resampling, which composite or resample without changing content, reduce image-level AUROC (99th-percentile score) by 1.6–3.0 points for the three-seed U-Nets and by at most 4.5 points for the other standard segmenters, but by 3.5–6.5 points for the marker-repaired models (one seed; the detector was not evaluated on these controls), and the PIX model raises the fewest false alarms on authentic images (12 % vs. 26 % of images with $>$0.1 % positive pixels at 0.5; three seeds).

TABLE II

MARKER PROBES ON 380 VALIDATION PAIRS WITH RESIDUE (STANDARD U-NET ROWS: MEAN OF THREE SEEDS; ALL OTHER ROWS: SEED 42). AP: PER-IMAGE AP UNDER BOX WITH 95 % GROUP-BOOTSTRAP CI FOR THE REPAIR CONTRAST; COVER: FRACTION OF THE BOX ABOVE τ^*; AUROC: FORGED-TYPE INPUT VS. AUTHENTIC, MAXIMUM SCORE.

	forged		repaired	Δ AP	auth.+marker	
Model	AP	Cover	AP	(repair)	Cover	AUROC
U-Net BOX	0.988	0.94	0.766	−0.222	0.80	0.903
U-Net PIX	0.747	0.54	0.500	−0.247	0.42	0.858
DeepLabV3+ BOX	0.982	0.92	0.807	−0.174	0.78	0.899
U-Net+aux PIX	0.771	0.59	0.597	−0.174	0.48	0.873
repaired BOX	0.939	0.89	0.958	+0.019	0.42	0.695
repaired PIX_{nm}	0.648	0.44	0.686	+0.038	0.11	0.592

C. The marker shortcut, and retraining without it

Fig. 1 defines the input probes; Table II reports them on the 380 validation pairs whose forgery carries marker residue (three-seed means for the standard U-Nets, seed 42 for all other rows). For the standard box U-Net, copying only the residue pixels back from the authentic image lowers per-image AP from 0.988 to 0.766 ($\Delta = -0.222$ [-0.256, -0.190]), and pasting only the residue onto the authentic image makes the model cover 80 % of the box at τ^* (AP 0.905). DeepLabV3+ and the pixel-trained U-Net show the same dependence (Δ -0.174 and -0.247). On the official test set the box model's AP is 0.972 on the 627 images with residue and 0.690 on the 185 without (0.722 on the 102 residue-free Gemini images), so the gap is not only a generator gap.

Retraining on marker-repaired images removes the dependence without destroying the benchmark score. The repaired box model loses nothing when the residue is removed at test time ($\Delta = +0.019$ [0.002, 0.036]); the residue alone now covers 42 % of the box instead of 80 % (11 % instead of 42 % for the pixel model); its test AUPRC is 0.939 (0.950 before), its AP on residue images falls from 0.972 to 0.920, while on residue-free Gemini images it is similar (0.756 vs. 0.722, overlapping CIs). The marker is thus a shortcut the models take when it is offered, not a prerequisite for high box scores. All repaired results use one seed.

D. Re-rendering versus content

Generators re-render the whole field, so inside the box many pixels change by less than the PIX threshold. Separating the two components on all validation pairs (Table III) shows that neither alone reproduces the response to a real forgery. For the standard box U-Net (three seeds) the mean probability inside the box falls from 0.86 (forged) to 0.24 (render-only) and 0.35 (content-only), and forged-vs-authentic AUROC from 0.971 to 0.637 and 0.733: the standard models respond most strongly to the conjunction of re-rendering, changed glyphs and marker. Removing the marker from training (one seed) shifts the models toward content: content-only AUROC rises to 0.872 for the box model and to 0.834 for the pixel model (0.681 before), while their render-only response stays lower

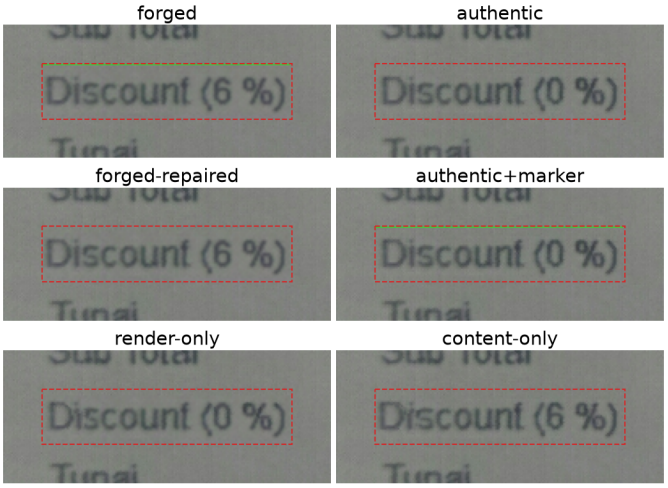


Fig. 1. Input probes on one validation pair (dashed: official box). The forged field carries a 1-px green residue of the generation marker on its top edge; *render-only* keeps the generator's re-rendering but restores the original digits; *content-only* carries the changed digits into the authentic image without the re-rendering or the marker.

TABLE III
RE-RENDERING VS. CONTENT (ALL VALIDATION PAIRS; STANDARD U-NET ROWS: MEAN OF THREE SEEDS, OTHER ROWS SEED 42): MEAN PROBABILITY INSIDE THE BOX AND IMAGE-LEVEL AUROC AGAINST AUTHENTIC IMAGES.

	mean p in box			AUROC vs. authentic		
Model	forged	render	content	forged	render	content
U-Net BOX	0.86	0.24	0.35	0.971	0.637	0.733
U-Net PIX	0.41	0.05	0.15	0.942	0.554	0.681
DeepLabV3+ BOX	0.87	0.28	0.42	0.972	0.644	0.755
repaired BOX	0.83	0.36	0.59	0.947	0.713	0.872
repaired PIX_{nm}	0.34	0.10	0.29	0.869	0.638	0.834

(0.713, 0.638). Content changes, not the generator's texture, become the main cue once the shortcut is gone.

E. Held-out generator (H3), now with three seeds

Training on Gemini only and testing on the Ideogram edits of validation and test (n=126 under BOX, 41 pairs under PIX) confirms our earlier single-seed result with three seeds (Table IV). Every Gemini-only model collapses on Ideogram; the BOX model falls from 0.909 to 0.412 per-image AP and the PIX model from 0.567 to 0.264 (both scored under BOX). The relative drop is smaller for pixel supervision, but its absolute Ideogram score is lower, and under its own ground truth the pixel model is no better on Ideogram than the box model (0.279 vs. 0.259). Pixel-change supervision does not, by itself, buy generator transfer. Because Ideogram forgeries almost never carry the marker (green residue in the box ring of 2.1 % of Ideogram vs. 85.9 % of Gemini forgeries), part of this gap may be a marker gap, which this design cannot separate: we did not train Gemini-only models on marker-repaired images.

TABLE IV
HELD-OUT GENERATOR: PER-IMAGE AP WITH 95 % GROUP-BOOTSTRAP CI, IDEOGRAM VS. SOURCE-MATCHED GEMINI (VAL+TEST, MEAN OVER THREE SEEDS).

Training	Scored	Ideogram	Gemini (matched)
All tools, BOX	BOX	0.629 [0.569, 0.692]	0.912 [0.896, 0.928]
Gemini only, BOX	BOX	0.412 [0.341, 0.481]	0.909 [0.892, 0.927]
Gemini only, PIX	BOX	0.264 [0.213, 0.313]	0.567 [0.550, 0.583]
Gemini only, BOX	PIX	0.259 [0.168, 0.350]	0.349 [0.322, 0.377]
Gemini only, PIX	PIX	0.279 [0.182, 0.382]	0.577 [0.546, 0.608]
Gemini only, GLYPH	PIX	0.182 [0.121, 0.246]	0.388 [0.363, 0.413]

F. Supervision variants, training length and robustness

Joint supervision (pixel mask plus an auxiliary box channel, seed 42) is the only variant that improves the pixel model on almost every axis we measured (the 89-pair test pixel AUPRC is unchanged, 0.650 vs. 0.654): validation AUPRC under PIX rises from 0.610 to 0.622 and under BOX from 0.579 to 0.647, test BOX AUPRC from 0.638 to 0.682, and the blur penalty shrinks from $-59\,\%$ to $-43\,\%$ (Table I; blur from the robustness runs). Image-level discrimination against authentic images is unchanged (0.897 vs. 0.892). With one seed these gains are suggestive, not established. *Twenty epochs* do not change the picture: BOX 0.926 (vs. 0.923), PIX 0.594 (vs. 0.610), GLYPH 0.645 (vs. 0.676); the concern that ten epochs were too few is not supported. *Test-time preprocessing.* Converting the input to grayscale is the most damaging operation we tested, costing 47–67 % of the AUPRC of every segmenter for which it was measured (seed-42 models, including the marker-repaired ones) and 59 % for the detector, so colour carries evidence beyond the green line. CLAHE changes AUPRC by at most 3 %. JPEG Q75, $0.75\times$ rescaling and σ=1 blur hurt pixel and glyph models more than box models (e.g. blur $-59\,\%$ vs. $-21\,\%$ for the U-Net), and the marker-repaired models are less robust to JPEG (-32 to $-50\,\%$) than their standard counterparts.

G. Box detection versus segmentation

A tiled YOLO11s trained on the same boxes is the strongest model in the study *under the box metric* and is no better than the box U-Net under the pixel metric (Tables I and V). Over three seeds it reaches 0.965 validation and 0.969 test AUPRC under BOX (U-Net 0.923 / 0.950), test F1@τ^* 0.946 and IoU 0.918 (U-Net 0.846 / 0.789) and box AP@0.5 0.880 (U-Net 0.873 for seeds 42/43; 0.391 for seed 44, whose calibrated τ^*=0.95 fragments the predicted region), while its PIX AUPRC is 0.368 (U-Net BOX model 0.374). The detector depends far less on the marker: its test AP on the 185 residue-free images is 0.878 (U-Net 0.690) and 0.886 on residue-free Gemini images (0.722). It also transfers better across generators: trained on all tools it reaches 0.872 AP on Ideogram (U-Net 0.629; both three seeds), and trained on Gemini only it keeps 0.752 (one seed; U-Net 0.412 over three seeds; Gemini 0.938 vs. 0.909). The probes (seed 42 for the detector) suggest why: removing the residue costs the

TABLE V
BOX DETECTOR VS. SEGMENTERS ON THE OFFICIAL TEST SET (812 IMAGES, BOX). AUPRC: POOLED PIXEL AUPRC (BOX PAINTED WITH ITS CONFIDENCE); AP: PER-IMAGE AP ON IMAGES WITHOUT / WITH MARKER RESIDUE; AP_{50}: BOX AP@0.5 (ALL IMAGES). SEEDS: YOLO11S AND U-NET 3, OTHERS 1. MS: FULL-IMAGE INFERENCE ON AN IDLE RTX 5080.

Model	AUPRC	F1@τ^*	AP no-mk	AP mk	AP_{50}	ms
U-Net BOX (3)	0.950	0.846	0.690	0.972	0.873†	81
U-Net PIX (3)	0.638	0.616	0.503	0.639	0.107	81
DeepLabV3+ BOX	0.952	0.833	0.702	0.962	0.833	122
U-Net BOX, repaired	0.939	0.797	0.701	0.920	0.805	81
YOLO11s tiles (3)	**0.969**	**0.946**	0.878	0.954	0.880	98
YOLO11n tiles	0.963	0.926	0.812	0.946	0.871	81
YOLO11m tiles	**0.969**	**0.947**	**0.882**	0.954	0.898	76
YOLO11s full 1280	0.932	0.909	0.756	0.930	**0.960**	303

†seeds 42/43; seed 44 (τ^*=0.95) 0.391. Zero-shot TruFor: F1@0.5 0.403.

detector only 0.062 AP (U-Net 0.222), whereas the *render-only* input, which keeps the generator's re-rendering but no changed glyph, is still separated from authentic images with AUROC 0.865 (U-Net 0.637). The detector has learned what a regenerated field looks like, a cue that is shared by both generators but says nothing about which pixels changed.

The ablations (seed 42) are only partly as expected. Resizing whole images to 1280 instead of tiling at native resolution costs 0.037 test AUPRC and 0.122 AP on residue-free images, but *raises* box AP@0.5 from 0.88 to 0.96 (0.93 on fields below 1 % of the image): a plausible cause is that fields cut at tile borders yield partial boxes that cross-tile NMS does not merge, although the untiled detector also emits about five boxes per image, so the mechanism is not established; tiling helps coverage but not box precision here; model size matters little (n/s/m: 0.963/0.969/0.969 test AUPRC); Ultralytics' default augmentation is as good as the conservative profile (0.971), contrary to our expectation that mosaic and scale jitter would destroy forensic traces; one extra authentic tile per image changes nothing (0.971). The detector is insensitive to blur, CLAHE and rescaling ($\leq$1.4 % AUPRC) but loses 49 % under JPEG Q75 and 59 % under grayscale. Contrary to our expectation, detection was not cheaper in our implementation: full-image inference takes 76–98 ms for the tiled n/s/m detectors, 303 ms for the 1280 detector, 81 ms for the U-Net and 122 ms for DeepLabV3+ (mean 3.1 MP per image), because per-tile pre- and post-processing dominates.

VI. DISCUSSION

What the box label rewards. Three independent lines of evidence point the same way. The same predictions lose 0.55 AUPRC when scored against the pixels that actually changed (Sec. V-A); the box model is indifferent to blurring the box interior but not its rim; and a box detector trained on the same labels reaches 0.969 test AUPRC under the box metric, more than any segmenter, while scoring 0.368 under the pixel-change truth. A benchmark that scores localization against the edited field's box therefore measures *field detection*. That is a useful capability for triage, and box detection delivers it with the highest box scores, but it should not be reported as pixel-level tamper localization.

The marker is a shortcut, not the whole story. Pasting only the residual green line onto an authentic image is enough for the standard box model to find the field, and its test AP drops from 0.97 on images that carry the residue to 0.69 on images that do not. Yet a box model retrained on marker-repaired images (one seed) keeps 0.916 validation AUPRC (0.923 before) and scores similarly on residue-free Gemini images (0.756 vs. 0.722, overlapping CIs). The residue is used when present and replaced by other cues when removed, which means the headline AIForge-Doc numbers overstate the difficulty-adjusted performance on marker-bearing images, but do not consist of the marker alone. Since the residue is present in 86 % of Gemini and 2 % of Ideogram forgeries, held-out-generator results on v1 confound generator and marker; repairing the marker is a prerequisite for any generator-transfer claim on this dataset.

Recommendations for document-forgery benchmarks. (i) Release the authentic source with every training forgery, as AIForge-Doc does, and report a pixel-change metric next to the box metric. (ii) Remove generation-pipeline annotations (markers, crop borders) from the released images and audit residue with a simple colour test. (iii) Report box-level AP separately from pixel metrics, with a box detector as a baseline. (iv) Report leakage-free test subsets and group-aware confidence intervals.

Limitations. The pixel-change ground truth is derived automatically (ε=8 with morphology); its agreement with a vision-language check is high (69 OK, 26 partial, 1 wrong of 96 pairs), but no human annotation was performed. The second architecture, the marker-repaired and joint-loss models, the 20-epoch runs and the YOLO ablations use one seed; only the main U-Net grid, the held-out-generator U-Nets and the main YOLO model use three. Ideogram contributes 126 held-out images. TruFor could not be fine-tuned because its training code is not released, so learned in-domain models are compared with zero-shot references only. Image-level detection is measured on validation pairs because the test authentic set is not paired. We evaluate one dataset; AIForge-Doc v2 [3] uses the same box annotation and is the natural next test.

VII. CONCLUSION

On AI-inpainted financial documents, the official box masks train and reward detectors of the edited field, not localizers of the changed pixels: changing only the training mask reverses the model ranking, the effect replicates across decoders, and a tiled box detector is the best model under the box metric and transfers best across generators, yet localizes changed pixels no better than the box-trained U-Net. A residual edit marker in most Gemini forgeries is exploited when present; retraining without it preserves most of the box score but removes the marker-specific advantage. Pixel-change supervision gives the most faithful masks and the fewest false alarms, but it does not close the generator gap (joint supervision was not tested on the held-out generator). We recommend reporting pixel-change

and box metrics side by side, auditing generation residue, and treating held-out-generator results on marker-bearing data with caution.

REFERENCES

[1] J. Wu, Y. Zhou, M. Xu, Z. Liang, S. Ren, J. Xue, M. Yang, S. Chen, and J. Huan, "AIForge-Doc: A benchmark for detecting AI-forged tampering in financial and form documents," 2026. [Online]. Available: https://arxiv.org/abs/2602.20569

[2] F. Guillaro, D. Cozzolino, A. Sud, N. Dufour, and L. Verdoliva, "TruFor: Leveraging all-round clues for trustworthy image forgery detection and localization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 20 606–20 615.

[3] J. Wu, Y. Zhou, D. T. Ng, X. Shen, K. Zewde, A. Raj, T. Duong, and S. Ren, "When the forger is the judge: GPT-Image-2 cannot recognize its own faked documents," 2026. [Online]. Available: https://arxiv.org/abs/2604.25213

[4] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.

[5] X. Shang, Y. Tang, J. Cui, A. Elhagry, S. K. Al Khatib, S. M. Bsharat, J. Liu, X. Zhao, J.-H. Xue, H. Li, S. Khan, and Z. Shen, "From masks to pixels and meaning: A new taxonomy, benchmark, and metrics for VLM image tampering," 2026. [Online]. Available: https://arxiv.org/abs/2603.20193

[6] C. Qu, C. Liu, Y. Liu, X. Chen, D. Peng, F. Guo, and L. Jin, "Towards robust tampered text detection in document image: New dataset and new solution," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 5937–5946.

[7] Z. Chen, S. Chen, T. Yao, K. Sun, S. Ding, X. Lin, L. Cao, and R. Ji, "Enhancing tampered text detection through frequency feature fusion and decomposition," in *Computer Vision – ECCV 2024*, ser. Lecture Notes in Computer Science, vol. 15091. Springer, 2024, pp. 200–217.

[8] K. Wong, J. Zhou, H. Wu, Y.-W. Si, and J. Zhou, "ADCD-Net: Robust document image forgery localization via adaptive DCT feature and hierarchical content disentanglement," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 19 280–19 289.

[9] M. Dhouib, Y. Zhu, S. Vanier, and A. Shabou, "Efficient document tampering localization with multi-level discrepancy features and unified DCT-quantization embedding," 2026. [Online]. Available: https://arxiv.org/abs/2606.22285

[10] K. Ronfleux-Corail, G. Bernard, M. Coustaty, and N. Sidère, "DocQT: Improving document forgery localization robustness via diverse JPEG quantization tables," 2026. [Online]. Available: https://arxiv.org/abs/2605.19688

[11] X. Liao, S. Chen, J. Chen, T. Wang, and X. Li, "CTP-Net: Character texture perception network for document image forgery localization," 2023. [Online]. Available: https://arxiv.org/abs/2308.02158

[12] Y. Song, W. Jiang, X. Chai, Z. Gan, M. Zhou, and L. Chen, "Cross-attention based two-branch networks for document image forgery localization in the metaverse," *ACM Transactions on Multimedia Computing, Communications, and Applications*, vol. 21, no. 2, pp. 1–24, 2024.

[13] L. Dong, W. Liang, and R. Wang, "Robust text image tampering localization via forgery traces enhancement and multiscale attention," *IEEE Transactions on Consumer Electronics*, vol. 70, no. 1, pp. 3495–3507, 2024.

[14] S. Li, Y. Guo, S. Chen, B. Li, K. Lin, C. Chen, H. Li, T. Yao, and S. Ding, "DITL^2: Dual-stage invariance transfer learning for generalizable document image tampering localization," in *Proceedings of the 33rd ACM International Conference on Multimedia (MM)*, 2025, pp. 82–91.

[15] Y. Wang, H. Xie, M. Xing, J. Wang, S. Zhu, and Y. Zhang, "Detecting tampered scene text in the wild," in *Computer Vision – ECCV 2022*, ser. Lecture Notes in Computer Science. Springer, 2022, pp. 215–232.

[16] C. Qu, Y. Zhong, F. Guo, and L. Jin, "Revisiting tampered scene text detection in the era of generative AI," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 1, 2025, pp. 694–702.

[17] Y. Wang, B. Zhang, H. Xie, and Y. Zhang, "Tampered text detection via RGB and frequency relationship modeling," *Chinese Journal of Network and Information Security*, 2022, in Chinese.

[18] D. Luo, Y. Zhou, R. Yang, Y. Liu, X. Liu, J. Zeng, E. Zhang, B. Yang, Z. Huang, L. Jin, and X. Bai, "ICDAR 2023 competition on detecting tampered text in images," in *Document Analysis and Recognition – ICDAR 2023*, ser. Lecture Notes in Computer Science, vol. 14188. Springer, 2023, pp. 587–600.

[19] D. Luo, Y. Liu, R. Yang, X. Liu, J. Zeng, Y. Zhou, and X. Bai, "Toward real text manipulation detection: New dataset and new solution," *Pattern Recognition*, vol. 157, p. 110828, 2025.

[20] G. Xu, P. Zhang, and L. Jin, "Bridging the reality gap in tampered text detection: A human-crafted real-world dataset and a text-centric approach," *IEEE Transactions on Information Forensics and Security*, vol. 21, pp. 6498–6513, 2026.

[21] C. Qu, J. Liu, H. Chen, B. Yu, J. Liu, W. Wang, and L. Jin, "TextSleuth: Towards explainable tampered text detection," 2024. [Online]. Available: https://arxiv.org/abs/2412.14816

[22] C. Qu, Y. Zhong, J. Liu, X. Zhu, B. Yu, and L. Jin, "TextShield-R1: Reinforced reasoning for tampered text detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 10, 2026, pp. 8621–8629.

[23] F. Zeng, C. Miao, J. Huang, Z. Tan, S. Gong, X. Yu, Y. Wang, W. Yao, J. T. Zhou, J. Li, and Y. Yan, "DocShield: Towards AI document safety via evidence-grounded agentic reasoning," 2026. [Online]. Available: https://arxiv.org/abs/2604.02694

[24] J. Zhang, S. Gao, Z. Chen, Y. Liu, and Y. Zhou, "Open-set visual text forensics via sparse-constraint rectified flow," 2026. [Online]. Available: https://arxiv.org/abs/2608.02258

[25] GenText-Forensics Challenge Organizers, "GenText-Forensics: Challenge on explainable forensics and adversarial generation for text-centric images," ACM Multimedia 2026 Grand Challenge, 2026, benchmark: RealText-V2, https://huggingface.co/datasets/vankey/RealText-V2. Accessed 9 November 2026. [Online]. Available: https://gentext-forensics-acm-mm-2026.github.io/

[26] Z. Zhao, W. Xia, E. Wei, Y. Zhang, J. Mo, T. Zhang, Y. Dai, Z. Chen, Y. Tao, and S. Ren, "DOCFORGE-BENCH: A comprehensive 0-shot benchmark for document forgery detection and analysis," 2026. [Online]. Available: https://arxiv.org/abs/2603.01433

[27] Y. Zhang, S. Ren, A. Raj, E. Wei, D. Ng, A. Shen, J. Xue, Y. Zhang, and E. Marotta, "GPT4o-Receipt: A dataset and human study for AI-generated document forensics," 2026. [Online]. Available: https://arxiv.org/abs/2603.11442

[28] Z. Liang, K. Zewde, R. P. Singh, D. Patil, Z. Chen, J. Xue, Y. Yao, Y. Chen, Q. Liu, and S. Ren, "Can multi-modal (reasoning) LLMs detect document manipulation?" 2025. [Online]. Available: https://arxiv.org/abs/2508.11021

[29] Y. Zhang, C. Miao, M. Liao, T. Liu, X. Wang, T. Gong, Q. Chu, and N. Yu, "TextFake: Benchmarking AI-generated image detection on text-rich images," 2026. [Online]. Available: https://arxiv.org/abs/2606.01050

[30] A. Raj, Y. Zhang, K. Zewde, T. Duong, J. Gan, X. Shen, Y. Zhou, H. Guo, S. Zhang, and S. Ren, "ChatGPT images 2.5 on forgery tasks: Testing advertised improvements against known answers," 2026. [Online]. Available: https://arxiv.org/abs/2609.13617

[31] B. Helali, "Beyond zero-shot on AIForge-Doc v2: A trained detector and a process-twin control," Zenodo, software and results archive, version 1.0, 2026, code: https://github.com/bilLkarkariy/aiforge-doc-v2-process-twins.

[32] leongyiquan, "DocDet: Detecting AI-generated content in document images," GitHub repository, 2026, accessed 9 November 2026. [Online]. Available: https://github.com/leongyiquan/DocDet

[33] S. Smeu, D.-A. Boldisor, D. Oneata, and E. Oneata, "Circumventing shortcuts in audio-visual deepfake detection datasets with unsupervised learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 18 815–18 825.

[34] A. Yermakov, J. Cech, J. Matas, and M. Fritz, "Deepfake detection that generalizes across benchmarks," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026, pp. 773–783.

[35] C. Liu, T. Zhu, W. Zhou, and W. Zhao, "Frequency bias matters: Diving into robust and generalized deep image forgery detection," 2025. [Online]. Available: https://arxiv.org/abs/2511.19886

[36] H. Mareen, D. Karageorgiou, G. Van Wallendael, P. Lambert, and S. Papadopoulos, "TGIF: Text-guided inpainting forgery dataset," in *IEEE International Workshop on Information Forensics and Security (WIFS)*, 2024, pp. 1–6.

[37] E. Nebioglu, E. Bilgiç, and A. Popescu, "AI-generated image detectors overrely on global artifacts: Evidence from inpainting exchange," 2026. [Online]. Available: https://arxiv.org/abs/2602.00192

[38] P. Giakoumoglou and S. Papadopoulos, "DiffusionPrint: Learning generative fingerprints for diffusion-based inpainting localization," 2026. [Online]. Available: https://arxiv.org/abs/2604.12443

[39] H. Mareen, D. Karageorgiou, P. Giakoumoglou, P. Lambert, S. Papadopoulos, and G. Van Wallendael, "TGIF2: Extended text-guided inpainting forgery dataset & benchmark," 2026. [Online]. Available: https://arxiv.org/abs/2603.28613

[40] S. Pandolfini, L. Pellegrini, M. Ferrara, and D. Maltoni, "Detecting localized deepfakes: How well do synthetic image detectors handle inpainting?" 2025. [Online]. Available: https://arxiv.org/abs/2512.16688

[41] L. Cai, H. Wang, J. Ji, Y. Zhoumen, S. Chen, T. Yao, and X. Sun, "Zooming in on fakes: A novel dataset for localized AI-generated image detection with forgery amplification approach," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 4, 2026, pp. 2534–2542.

[42] S. Aoudi, S. A. P., N. Al Barghuthi, and O. Alamir, "Beyond accuracy: Auditing image forgery localization via scene-disjoint evaluation and XAI faithfulness," *IEEE Access*, vol. 14, pp. 53 101–53 115, 2026.

[43] B. Dou, Z. Wang, H. Wang, G. Cai, W. Tan, C. Si, L. Guo, and Y. Lyu, "GIFTBench: Diagnosing generalization in image forgery localization and informing model design," 2026. [Online]. Available: https://arxiv.org/abs/2610.01778

[44] J. Dai, K. He, and J. Sun, "BoxSup: Exploiting bounding boxes to supervise convolutional networks for semantic segmentation," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015, pp. 1635–1643.

[45] Z. Tian, C. Shen, X. Wang, and H. Chen, "BoxInst: High-performance instance segmentation with box annotations," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 5439–5448.

[46] L. Maier-Hein, A. Reinke, P. Godau, M. D. Tizabi, F. Buettner *et al.*, "Metrics reloaded: recommendations for image analysis validation," *Nature Methods*, vol. 21, no. 2, pp. 195–212, 2024.

[47] P. Zhou, X. Han, V. I. Morariu, and L. S. Davis, "Learning rich features for image manipulation detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 1053–1061.

[48] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.

[49] G. Jocher and J. Qiu, "Ultralytics YOLO11," Software, version 11.0.0, AGPL-3.0, 2024. [Online]. Available: https://github.com/ultralytics/ultralytics

[50] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Computer Vision – ECCV 2014*, ser. Lecture Notes in Computer Science. Springer, 2014, pp. 740–755.

[51] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, ser. Lecture Notes in Computer Science. Springer, 2015, pp. 234–241.

[52] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Computer Vision – ECCV 2018*, ser. Lecture Notes in Computer Science. Springer, 2018, pp. 833–851.

[53] G. Cheng, X. Yuan, X. Yao, K. Yan, Q. Zeng, X. Xie, and J. Han, "Towards large-scale small object detection: Survey and benchmarks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023.

[54] F. C. Akyon, S. O. Altinuc, and A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," in *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, 2022, pp. 966–970.

[55] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.

[56] GuangJian Team, "Venus-DeFakerOne: Unified fake image detection & localization," 2026. [Online]. Available: https://arxiv.org/abs/2605.14091

[57] J. Sauvola and M. Pietikäinen, "Adaptive document image binarization," *Pattern Recognition*, vol. 33, no. 2, pp. 225–236, 2000.

[58] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[59] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Fourth International Conference on 3D Vision (3DV)*, 2016, pp. 565–571.

[60] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[61] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, 1979.

[62] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.